NYCMOS: Selectable Homodyne/Sliding IF Downconversion Mixer

Team Lead: Demian Marin - demianmarin03@gmail.com

Lei Chen - leichen591238@gmail.com

Charbel El Haddad - charbel.elhad@gmail.com

Update

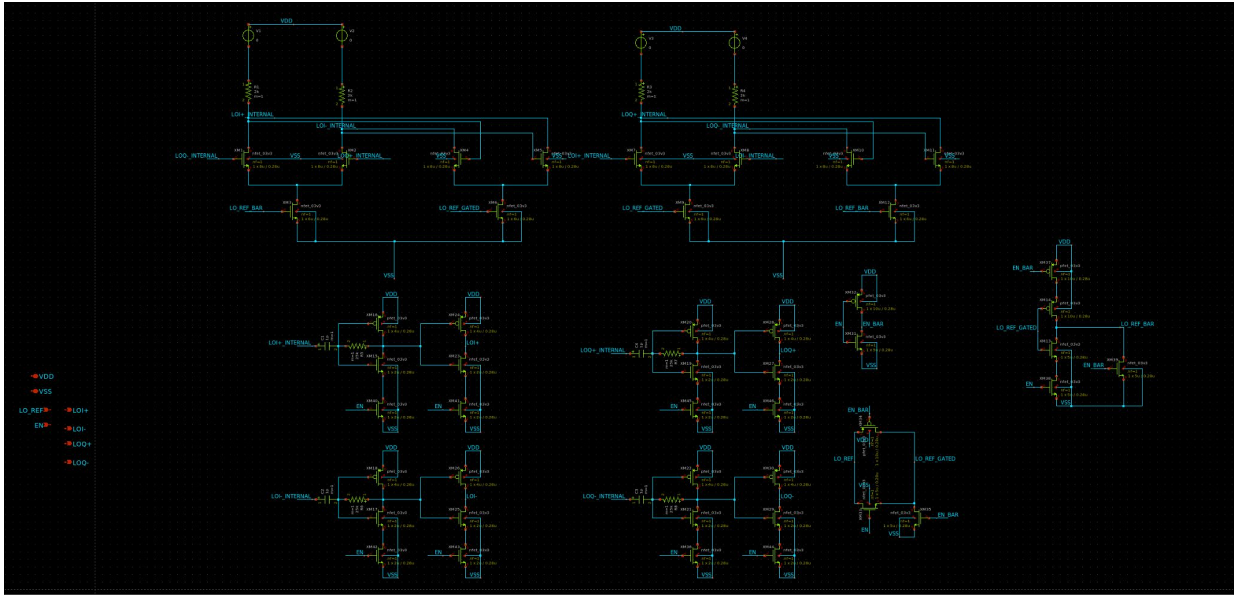
- %2 Frequency Divider Complete
- New block diagram
- Updated specs for the chip
- Potential experiments
- Task tracker

LO

- To generate quadrature LO signals, use reference LO at 2x the operating frequency, and divide it by 2 to get LOI and LOQ at the correct operating frequency
- Use flip-flop divider to avoid inductors which can be hard to design on chip
- PLL_MOSBIUS claims to operate from 5 MHz to 150 MHz, and Si535x chip operates up to 200 MHz.
- Set target operating frequency at 50-75 MHz given clock constraints.

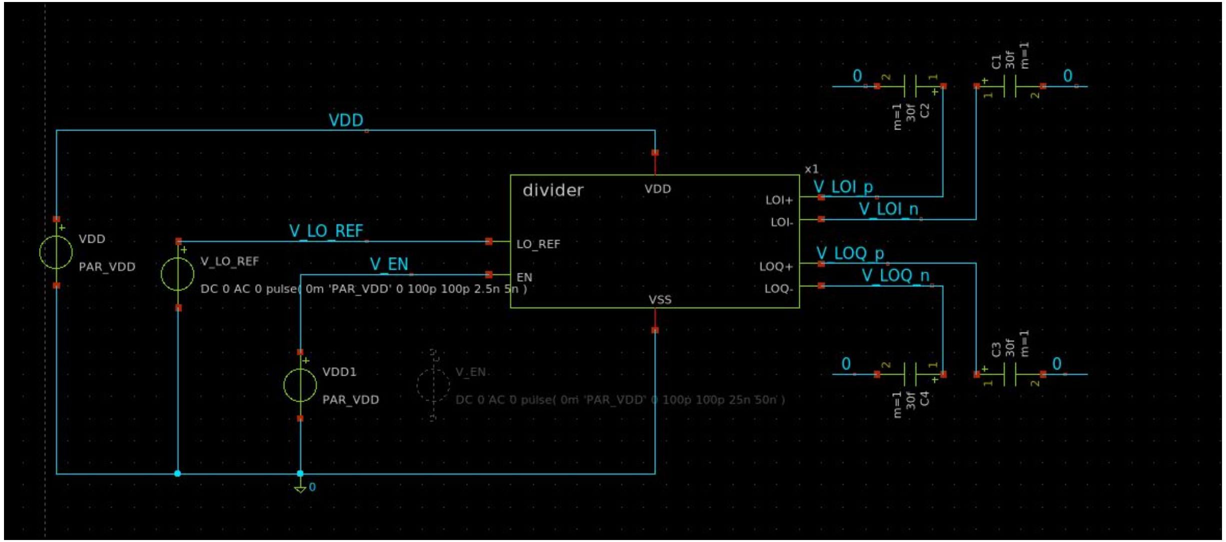
Divider

Flip flop based Divider with output buffers for differential square wave generation



Divider Testing

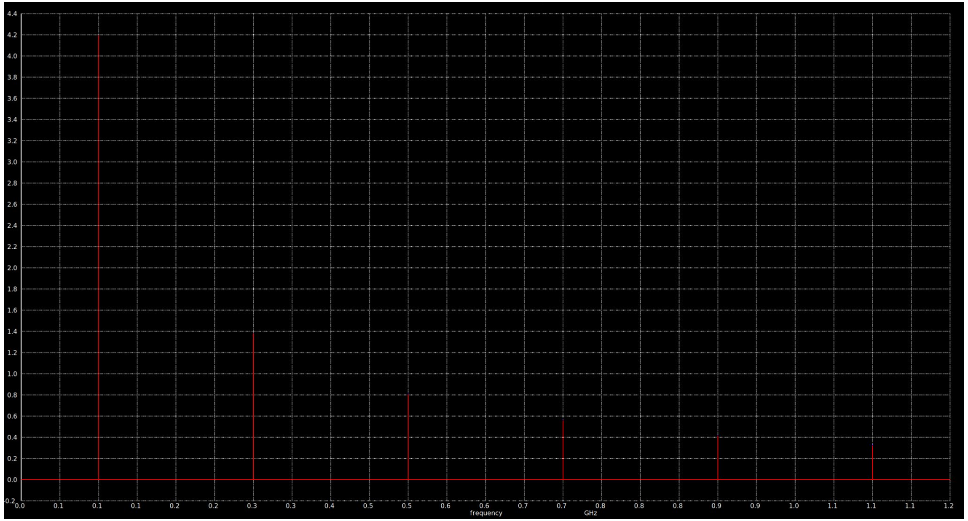
Testbench with 200 MHz 0-3.3V square wave

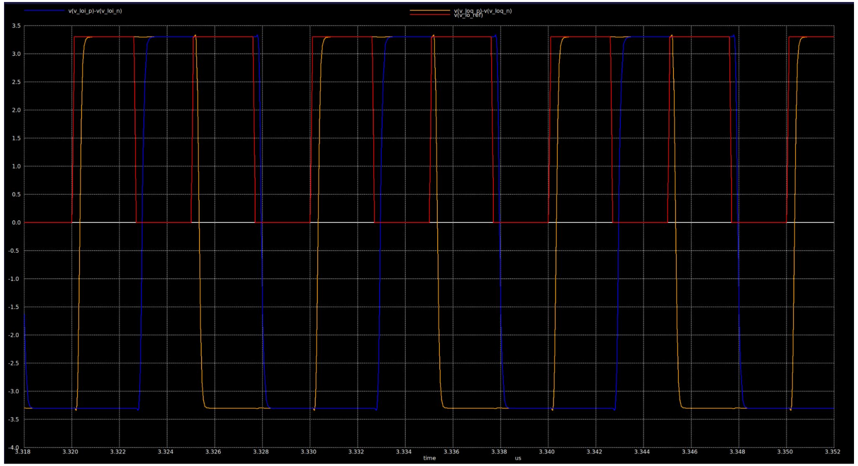


Divider Results

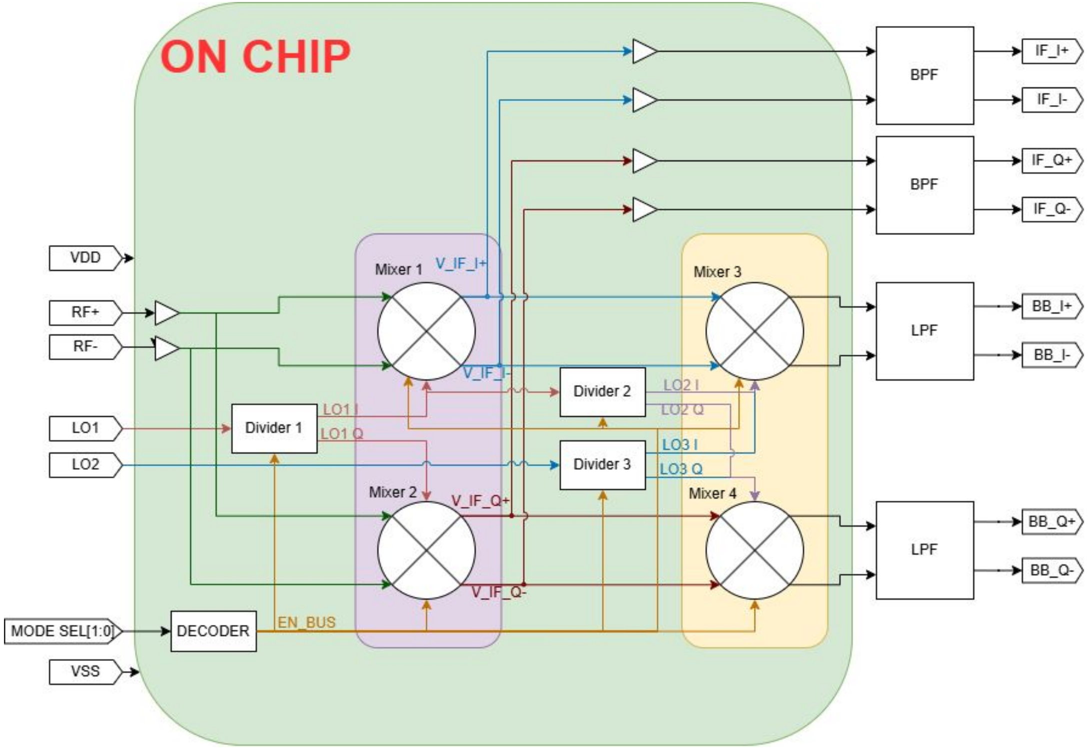
Spectrum shows fundamental at 100 MHz

Transient shows around 90 degree phase shift





Updated Block Diagram



- Removed muxes, instead we have a central EN_BUS that enables/disables sets of mixers and dividers based on user configuration
- Filters are offchip so that students are able to set their own filter bandwidths

Updated Specs

Parameter	Target
Power Supply	3.3V
Mixer Operating Frequency	25MHz-75MHz
Conversion Gain	Adjustable, 10-30 dB depending on current source on output amplifier
Noise Figure	10 dB? Will optimize for this after we get the mixers to work
IIP3	No target for now. Will measure and report

Possible Experiments

- Modulation Schemes are not specified. Students have access to 2 pairs of mixers for homodyne, heterodyne, sliding IF, and 2 stage IF topologies.
- Homodyne - Enable Mixers 1,2, Divider 1. Disable Mixers 3,4, Dividers 2,3. Unfiltered baseband output is exposed at IF_I+, IF_I-, IF_Q+, IF_Q-.
- Heterodyne - Enable Mixers 1,2, Divider 1. Disable Mixers 3,4, Dividers 2,3. Set LO frequency as desired based on IF frequency. Unfiltered baseband output exposed at IF_I+, IF_I-, IF_Q+, IF_Q-.
- Sliding IF - Enable Mixers 1,2,3,4, Divider 1,2. Disable Divider 3. Set LO to ⅔ of F_rf, baseband output at BB_I+, BB_I-, BB_Q+, BB_Q-.
- 2 Stage Heterodyne - Enable Mixers 1,2,3,4, Dividers 1,3. Disable Divider 2. Set LO1 and LO2 to desired frequency based on IF2 given F_rf.
- Since modulation schemes are not specified for the experiments, we don't have the channel width, and so it is up to the students to design LPF/BPF for their modulation scheme's channel width and IF.

Experiment Example: AM Demodulation with Image and Sliding IF Receiver

- Generate AM signal with 50kHz channel bandwidth (audio signal) modulated to 25 MHz with a function generator or some other means.
- Set the chip to 2 stage Heterodyne mode without quadrature mixers (Enable Mixers 1,3, Divider 1,3. Disable Divider 2.)
- Set LO frequency to 25MHz * ⅔ = 16.67 MHz. Generate LO via RF_PLL MOSBIUS chip or some other clock.
- Build a LPF with 25 KHz bandwidth because 50 kHz channel width is now centered at DC.
- Observe demodulated AM signal.
- Add a blocker at the image frequency 25 - (2*16.67 MHz) = -8.33 MHz
- The blocker at -8.33 MHz will be converted to 8.33 MHz with the LO of 16.67. This image is then converted by the second mixer stage to baseband, corrupting the signal.

Task Tracking Sheet

Sheet:
https://docs.google.com/spreadsheets/d/1Od1J7DOz2SekQrFCoG0DU00RiTUWZIC1DUhUMYMRFoQ/edit?usp=sharing